

Shuigen Liu

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Positions

- 2026 – Now** **Postdoctoral Scholar**, The Pennsylvania State University
Mentor: Prof. John Harlim
- 2025 – 2026** **Research Fellow**, National University of Singapore
- 2024 Fall** **Visiting student**, Massachusetts Institute of Technology

Education

- 2021 – 2025** **Ph.D. in Mathematics**, National University of Singapore
Advisors: Prof. Xin T. Tong and Prof. Weizhu Bao
- 2017 – 2021** **B.S. in Mathematics**, Peking University

Research Interests

My research lies at the intersection of computational mathematics, applied probability and statistics. I focus on analysis and development of algorithms for high-dimensional sampling and uncertainty quantification, with applications in Bayesian inference, inverse problems and generative modeling. I am particularly interested in localization methods for sampling. I also work on ensemble Kalman methods for high-dimensional inference and data assimilation.

Keywords: Sampling, uncertainty quantification, localization, dimension reduction, data assimilation.

Publications and Preprints

Preprints

- [6] Georg A. Gottwald, **Shuigen Liu**, Youssef Marzouk, Sebastian Reich, and Xin T. Tong, *Localized diffusion models*, arXiv:2505.04417, submitted to Bernoulli (2026).
- [5] Tiangan Cui, **Shuigen Liu**, and Xin T. Tong, *Stein's method for marginals on large graphical models*, arXiv:2410.11771, submitted to Bernoulli (2026).

Published/accepted

- [4] **Shuigen Liu**, and Xin T. Tong, *Spectral Gap of Metropolis Algorithms for Non-smooth Distributions under Isoperimetry*, Jpn. J. Stat. Data Sci. (2026).
- [3] Ningxin Liu, **Shuigen Liu**, Xin T. Tong, and Lijian Jiang, *Estimate of Koopman modes and eigenvalues with Kalman filter*, SIAM J. Sci. Comput., 48 (2026), pp. A512–A539.
- [2] Rafael Flock, **Shuigen Liu**, Yiqiu Dong, and Xin T. Tong, *Local MALA-within-Gibbs for Bayesian image deblurring with total variation prior*, SIAM J. Sci. Comput., 47 (2025), pp. A2127–A2153.
- [1] **Shuigen Liu**, Sebastian Reich and Xin T. Tong, *Dropout ensemble Kalman inversion for high dimensional inverse problems*, SIAM J. Numer. Anal., 63 (2025), pp. 685–715.

Teaching

The Pennsylvania State University

Lecturer Calculus With Analytic Geometry I, Fall 2026

National University of Singapore

TA Finite Element Method, Spring 2025

TA Essential Data Analytics Tools: Numerical Computation, Spring 2024

TA Calculus for Computing, Spring 2023 and Fall 2023

TA Essential Data Analytics Tools: Convex Optimization, Spring 2023

Awards

2024 Faculty of Science PhD Conference Award, National University of Singapore

2024 Overseas Research Immersion Award, National University of Singapore

2021 President's Graduate Fellowship, National University of Singapore

2021 Excellent Undergraduate, Peking University

Service

Reviewer for: SIAM Journal on Numerical Analysis, SIAM/ASA Journal on Uncertainty Quantification, Journal of Machine Learning Research, Journal of the Royal Statistical Society Series B, Statistics and Computing, Japanese Journal of Statistics and Data Science

Conferences

Jul 2025 Efficient Sampling Algorithms for Complex Models, IMS NUS, Singapore.

Jun 2025 MATRIX Research Program: Bayesian Learning of Very High-Dimensional Physical Process Models, MATRIX, Melbourne, Australia.

May 2025 2025 STU-NUS PhD Forum, Shanghai Jiao Tong University, Shanghai, China.

Oct 2024 FDS Conference: Recent Advances and Future Directions for Sampling, Yale University, New Haven, U.S.

Feb 2024 2024 SIAM Conference on Uncertainty Quantification (UQ24), Trieste, Italy.

Aug 2023 2023 International Congress on Industrial and Applied Mathematics (ICIAM 2023), Waseda University, Tokyo, Japan.

Jun 2023 Foundations of Computational Mathematics 2023 Conference (FoCM 2023), Sorbonne University, Paris, France.

Jul 2022 2022 International Conference on Scientific Computation and Differential Equations (SciCADE 2022), University of Iceland, Reykjavík, Iceland.